A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 14

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1 LEO:** high resolution and coverage of the whole Earth (including the poles) allows detailed mapping of ice area / thickness and forest cover **1** But each area is only revisited every 15 days so they cannot follow rapidly changing weather **1 Geostationary:** constant view of the same region allows continuous monitoring of cloud / storm movements, though the poles are not visible and detail is lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1** Wind velocity **1** Ocean currents / sea surface temperature **1** Wave height **1** Ice thickness **1** Vegetation cover **1** Sea level **1** Atmospheric temperature / composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Data for a past year (eg 1900) are fed in and the model's prediction for a later year (eg 2000) is compared with the real outcome **1** If the prediction matches, the model can be trusted (with caution) – it is continually modified as more data are collected and differences analysed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Models need data from all regions / oceans / altitudes – gaps (eg oceans, polar regions, past periods) introduce uncertainty into predictions **1** Unrepresentative data (eg urban heat islands, short time series) would bias the model – remote sensing helps fill gaps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 102; AQA spec 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Nuclei of large atoms (uranium-235, plutonium-239) split when hit by a neutron **1** A small amount of mass is converted to a large amount of energy ($E = mc^2$) and more neutrons are released, causing a chain reaction (controlled by control rods / moderator) **1** The heat boils water to steam which drives turbines and generators **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **control rods** → absorb neutrons to control the rate of fission • **moderator** → slows the neutrons so that fission continues **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Output changes slowly (days) so it cannot follow rapid fluctuations in demand **1** Very high construction cost but low fuel cost means it is most economic running continuously at full output **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Only 30 t of fuel a year (very high energy density) so no need for railways / ports to deliver millions of tonnes of coal **1** Still need condenser cooling water (coast / large lake) and a stable site **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
02.4	1 mark for the point below. **1** Uranium must be mined, purified, concentrated and processed into fuel using (fossil) energy – reactor construction and decommissioning have embodied energy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life • **embodied energy** → all the energy used to make and transport a product **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1

Do not write outside the box

Question	Marking guidance	Marks
02.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** No commercial reactor has been fully decommissioned so costs are unknown and often exceed estimates **1** Large projects often overrun original cost estimates **1** Long-term waste disposal costs uncertain **1** State subsidies may hide costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 191-193; AQA spec 3.3.4.1.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Cost increases as the percentage prevented increases **1** **The relationship is not linear:** cost rises slowly at first then very steeply / exponentially above about 80-90% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16.0 - 5.0 = 11.0 (£ million) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (£ million) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** **ALARA:** emissions should be As Low As Reasonably Achievable **1** It is rarely practical / affordable to reduce emissions to zero – low-level emissions may have acceptable impacts so an acceptable emission level is set **1** **BATNEEC:** Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive **1** The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Industry / city may have expanded so there are more pollutant sources and total emissions have risen **1** Better understanding of the health / environmental effects so stricter standards are set **1** Improved technology becomes available (BATNEEC changes) / cost of better control falls **1** Pollutant may have accumulated (persistent) so background levels are higher **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chemicals cause the oil molecules to join together / solidify **1** Producing more solid material that can be collected / lifted more easily than liquid oil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Air contains about 210 cm^3 of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm^3 per litre (less than 3% of the amount in air) **1** So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen – many aquatic organisms are near the lower end of their range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water • **range of tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 5 points below. **1** Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated **1** Same slope / gradient – simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time **1** Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it **1** Repeat each treatment several times and calculate means (and standard deviation) **1** Plot soil loss against cover – statistical test (Spearman's rank / t-test) for significance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean • **Spearman** → a test for correlation between two sets of ranked data • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	5
05.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Rainfall intensity / raindrop size / height **1** Soil texture, compaction, organic matter, depth **1** Permeability of the material beneath **1** Gradient **1** Root binding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Permeability** → how easily water passes through / lets water through **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** All variables except the independent variable can be standardised in the laboratory – in the field uncontrollable variables (weather) must be measured instead **1** Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **valid** → actually measures what it is meant to measure	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - tiny pores between clay particles allow capillary rise – none in sand	1
06.2	**1 mark each** 1 mark for each of the 3 points below. **1 Hygroscopic:** held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants **1 Capillary:** held in pores by surface tension – does not drain but can be absorbed by plants **1 Gravitational:** drains freely through large pores – available to plants only briefly as it passes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.3	1 mark for the point below. **1** Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **peds** → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (52 - 40) / 40 x 100 **1** = 30% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Sulfur dioxide came mainly from burning coal – domestic coal fires were controlled by the Clean Air Acts and coal-fired power stations have closed or been fitted with flue gas desulfurisation **1** Low-sulfur fuels are now used, and sulfur can be removed from the fuel before combustion **1** Nitrogen oxides are produced whenever any fuel is burned in air at high temperature, so they are not removed by changing the fuel – road traffic in the city centre has continued to grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	3
07.3	**1 mark each** 1 mark for each of the 3 points below. **1 Electrostatic precipitator:** the flue gases pass charged wires that give the particulates an electric charge, so they are attracted to oppositely charged collecting plates and can be shaken off and removed **1 Catalytic converter:** a platinum / rhodium catalyst converts oxides of nitrogen, carbon monoxide and unburnt hydrocarbons into nitrogen, carbon dioxide and water **1 Low-NOx burner:** the fuel is burned at a lower temperature, or in stages, so less of the nitrogen and oxygen in the air combines to form oxides of nitrogen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Electrostatic precipitator** → charged plates that attract and collect particulates • **particulates** → fine solid particles / smoke and dust • **Catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** Ground-level ozone is a secondary pollutant: it is not emitted directly but formed in the air by sunlight acting on oxides of nitrogen and volatile organic compounds **1** Nitrogen dioxide is split by ultraviolet light to give nitrogen monoxide and an oxygen atom, which joins an oxygen molecule to make ozone **1** With less NOx there is less of the raw material for these photochemical reactions, so less ozone and fewer PANs are produced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • ground-level ozone described as protecting us from UV - that is stratospheric ozone; at ground level it is a pollutant	3

Do not write outside the box

Question	Marking guidance	Marks
07.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Lichens differ in their sensitivity to sulfur dioxide, so the species present show the average pollution level over a long period rather than at one instant **1** They need no power supply or expensive equipment and can be surveyed at many sites, giving good spatial coverage cheaply **1** They integrate the effects of pollution over years, so a short pollution episode may not show up **1** They indicate sulfur dioxide well but say little about nitrogen dioxide, ozone or particulates, which are now the city's main problems **1** Other factors affect them too, such as the pH and texture of the bark or stone they grow on, shade and humidity, so a survey can be misleading unless these are standardised **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **indicator species** → a species whose presence or absence shows the conditions present • **particulates** → fine solid particles / smoke and dust **Do not credit** • 'lichens die where the air is polluted' without naming the pollutant - it is their sensitivity to sulfur dioxide that is used	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	
08.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Orientation:** south-facing windows (northern hemisphere) maximise passive solar gains – rooms needing warmth on the sunny side – fewer / smaller windows on the north side where losses are greatest – solar screens / brise soleil prevent summer overheating **2** Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost **2 Earth sheltering:** the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy **1** Limecrete has about half the embodied energy of concrete – rammed earth, straw bales and lime mortar are lower still **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **embodied energy** → all the energy used to make and transport a product	2

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area **1** Heat transfers from the outgoing to the incoming air so fresh air enters already warmed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2.	
09.1	**1 mark each** 1 mark for each of the 5 points below. **1** Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions **1** Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected – recirculation increases the metal concentration **1** Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) **1** Smelting needs chemical reduction at high temperatures – energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic **1** Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **Leachate** → polluted liquid draining out of waste **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves **1** The vegetation is harvested and incinerated **1** The concentrated metal in the ash is dissolved in acid and separated by electrolysis **OR** decontaminates polluted sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) **1** The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Organic nutrients (sewage, manure, silage effluent, paper mill / tannery / food processing effluent) are food for aerobic microbes whose respiration deoxygenates the water, killing fish and invertebrates – bacterial growths cover the bed, blocking light • Pathogens in sewage (cholera, typhoid, dysentery) spread via contaminated water – decay of organic matter releases inorganic nutrients • **Inorganic nutrients:** nitrates (fertiliser / manure runoff, leaching) and phosphates (sewage effluent, detergents, eroded soil) cause cultural eutrophication: algal blooms, shading of macrophytes, cyanobacterial toxins, night-time and end-of-summer deoxygenation • **Human health:** nitrites and methaemoglobinaemia – possible carcinogen via nitrosamines • **Control of organic wastes:** sewage treatment (pre-treatment, primary sedimentation removing >95% organic matter, aeration / activated sludge or trickling filters, anaerobic digestion of sludge with biogas) • **Control of nutrients:** tertiary phosphate stripping with iron(III) sulfate, denitrification in filter beds, micro-strainers / UV – Nitrate Vulnerable Zones, fertiliser timing / amount, low-solubility fertilisers, legumes, manure management – dredging sediments • **Evaluation:** sewage treatment has virtually eliminated deoxygenation and waterborne disease in developed countries but storm overflows and combined sewers still discharge untreated sewage – tertiary treatment is expensive so only used at sensitive sites – diffuse agricultural nitrate is hard to control and aquifers respond slowly – phosphate legacy in sediments • Judgement on point-source (sewage) control being more successful than diffuse-source (agriculture) control	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **eutrophication** → nutrient enrichment of water leading to algal growth • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (9 marks)	
	Organic wastes - sewage, manure, silage effluent, paper-mill and food-processing effluent - are food for aerobic bacteria, whose respiration strips dissolved oxygen from the water so fish and invertebrates suffocate and bacterial growths smother the bed; sewage also carries pathogens such as cholera and typhoid. Inorganic nutrients act differently: nitrate leached from fertiliser and manure and phosphate from sewage, detergents and eroded soil cause cultural eutrophication, in which algal blooms shade macrophytes, cyanobacteria release toxins and night-time respiration and decay deoxygenate the water. Nitrate in drinking water can cause methaemoglobinaemia and may form carcinogenic nitrosamines.	
	Point sources are controlled by sewage treatment: screening and grit removal, primary sedimentation removing most organic solids, aeration with activated sludge or trickling filters, and anaerobic digestion of sludge yielding biogas. Nutrients need tertiary treatment - phosphate stripping with iron(III) sulfate, denitrifying filter beds, micro-strainers and UV. Diffuse agricultural sources are managed by Nitrate Vulnerable Zones, limits on the timing and amount of fertiliser, low-solubility fertilisers, legumes, manure management and buffer strips.	
	Sewage treatment has virtually eliminated river deoxygenation and waterborne disease in developed countries, but storm overflows on combined sewers still discharge untreated sewage, and tertiary treatment is expensive so it is used only where eutrophication is a risk. Diffuse nitrate from farmland is much harder to control because it comes from thousands of sources and depends on farmers' behaviour and the weather. Control of point sources has therefore been far more successful than control of diffuse ones.	
	Weaker answer - Level 1 (3 marks)	
	Sewage in rivers uses up the oxygen because bacteria break it down, so fish die. Fertilisers wash into rivers and cause eutrophication, where algae grow and block the light and then the water runs out of oxygen. Sewage is treated in sewage works to clean it before it goes into rivers. Farmers should use less fertiliser. Sewage treatment works well but fertiliser is still a problem.	
	What makes the difference	
	• Level 1 has the outline of both mechanisms but Level 3 explains them precisely (aerobic respiration, shading, night-time respiration, pathogens, health effects) and names the pollutant sources.	
	• Level 3 describes the stages of sewage treatment and the specific nutrient controls; Level 1 says 'sewage works' and 'less fertiliser'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation gives reasons (storm overflows, cost of tertiary treatment, diffuse sources) for its point-vs-diffuse judgement; Level 1 asserts it.	
10.2	1 mark for the point below. **1** Algae / cyanobacteria absorb nutrients directly from the water (not the sediments) so they use dissolved nutrients more easily and grow rapidly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-272; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • **How properties predict behaviour:** dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • **Control based on properties:** persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones – systemic pesticides and seed coatings • **Heavy metals:** lead uses discontinued, storage at high pH to reduce solubility – mercury chemical form – radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays) – oil properties (floats, insoluble) exploited by booms and skimmers	25

Do not write outside the box

Question	Marking guidance	Marks
	– density in cyclone separators • **Principles:** CPA and CGM based on properties – emission location / timing – ALARA and BATNEEC – polluter pays – precautionary principle • **New technologies:** satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • **Evaluation:** prevention better than cure – limits of remediation – cost-effectiveness – contrast success of ozone (Montreal) with climate change – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	
	Model answer - Level 5 (22 marks)	
	Pollutants cause damage in proportion to how much reaches sensitive organisms and how harmful it is when it gets there. Both depend on properties, so knowing a pollutant's properties lets us predict where it will go and which control will work.	
	State of matter and density decide dispersal. Solid particulates settle near the source unless they are very small: PM10 and smaller stay suspended, travel far and penetrate lungs, so smoke is controlled at the point of release with electrostatic precipitators, cyclones and bag filters, and smokeless zones prevent its production. Gases disperse widely, and dense gases such as the hydrogen cyanide released at Bhopal hug the ground, so location and emission height matter. Persistence decides whether dilution is acceptable. Sewage and other biodegradable wastes break down, so treatment plus dilution in a river or the sea is sufficient; organochlorine pesticides, CFCs, heavy metals and long-lived radioisotopes are not broken down, so they travel, accumulate and must be contained, banned or replaced - DDT and CFCs were both controlled by substitution once their persistence was understood.	
	Solubility governs the pathway. Water-soluble nitrate leaches into rivers and aquifers, so it is controlled by nitrate vulnerable zones, fertiliser timing and buffer strips; liposoluble pollutants such as DDT and methyl mercury pass into fat and are biomagnified along food chains, as at Minamata, so even low concentrations in water are dangerous and the only effective control is to stop release. Adsorption on clay immobilises caesium and metals, which is why radioactive discharges are monitored in sediments and why activated carbon and polymer adsorbents are used deliberately in effluent treatment.	

Do not write outside the box

Question	Marking guidance	Marks
	Toxicity, specificity and mode of action set acceptable doses and identify species at risk. Neonicotinoids are toxic to insects but not vertebrates because they cannot cross the vertebrate blood-brain barrier, which explains both their popularity and their effect on bees; pyrethroids replaced organochlorines because they are less persistent. For mutagens and carcinogens such as ionising radiation there is no safe dose, so exposure is kept As Low As Reasonably Achievable with shielding, distance, time limits and monitoring. Reactivity explains secondary pollutants - sulfur dioxide oxidised to sulfuric acid, nitrogen oxides and hydrocarbons forming ozone and PANs in photochemical smog - so control targets the primary emissions with flue-gas desulfurisation and catalytic converters. Synergism between smoke and sulfur dioxide, or ozone and sulfur dioxide, means limits must consider pollutants together.	
	Properties interact with the environment: temperature inversions trap pollutants, currents disperse them, pH and oxygen alter solubility. Critical Pathway Analysis combines the two to predict where a discharge will concentrate and to focus monitoring, and Critical Group Monitoring checks the most exposed people. These principles have reduced smoke, sulfur dioxide, lead and organochlorines dramatically in developed countries. Their limits are pollutants with diffuse sources or that underpin economies - nitrate, carbon dioxide - where knowing the properties is not the problem; changing behaviour is. Even there, the precautionary principle applies the same logic: predict the properties of a new substance from similar ones and control it before release, because prevention is always cheaper than remediation.	
	What makes the difference	
	• Every paragraph is organised around a **property**, each linked to a prediction and a named control - the essay answers the question throughout rather than describing pollutants.	
	• Examples span the specification (smoke, acid rain, pesticides, heavy metals, radiation, nutrients, oil), with quantitative and named detail (Bhopal, Minamata, PM10, ALARA), which is what Level 5 breadth means.	
	• The final paragraph evaluates: where the approach has worked, where it has not and why, and reaches a justified conclusion.	
	• A Level 3 version of this essay would describe several pollutants and their effects accurately but only occasionally connect a property to a control, and would end without a judgement.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • **Renewable features:** intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal) – low energy density – location – low pollution but manufacturing impacts • Technologies and developments for each resource with examples – new technologies improving usability • **Storage and grid management:** pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving – the role of electricity as a secondary fuel • **Economics and politics:** development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution – sustainability principles - natural systems run on low-density renewable energy • **Judgement:** electricity largely yes	25
	– heat, aviation, shipping and high-temperature industry harder – transition period with nuclear / CCS *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

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